Failure-guided study design and separated evaluation tracks

GENERALIZATION TRACK

Frozen split

Train 2,510 | Validation 358 | Sealed test 717

Sealed test untouched

RQ1 Generative LoRA

Form learned: 351/358 structurally complete

Mapping not reliable: 0/358 exact

Development-validation behavior

RQ2 Train-only retrieval

Candidate reachability: L0 / L1 / L2 / L3

0 / 21 / 221 / 116

Genus signal improved

Complete-label coverage ceiling: 0/358

Availability separated from ranking

RQ3 Constraint diagnostics

Soft context → hard prefixes → oracle diagnostics

Constraint compliance ≠ knowledge acquisition

Oracles diagnose completion; not system performance

TASK-FORMULATION PIVOT

Do not regenerate authoritative terminology facts

COVERED-RESOURCE TRACK

Audited terminology knowledge base

3,593 terminology records

RQ4 Governed resolution chain

Exact main → verified alias → normalized match

Fuzzy retrieval → candidate only

Ambiguous / unsupported → abstain

Latin output only from unique immutable KB records

Frozen covered-resource benchmark

Automatic-resolution correctness: 1,400 / 1,403

Automatic coverage: 1,403 / 1,855

Review / abstention: 452 / 1,855

Unsafe automatic resolution: 3 / 1,855

Provenance complete: 1,403 / 1,403

Covered-resource performance ≠ unseen generalization

Abstain / human review

for ambiguity, unsupported input, or fuzzy candidates

Knowledge base = factual authority

Future LLM roles: query reformulation | candidate explanation | interactive clarification

NOT Latin-name factual generation